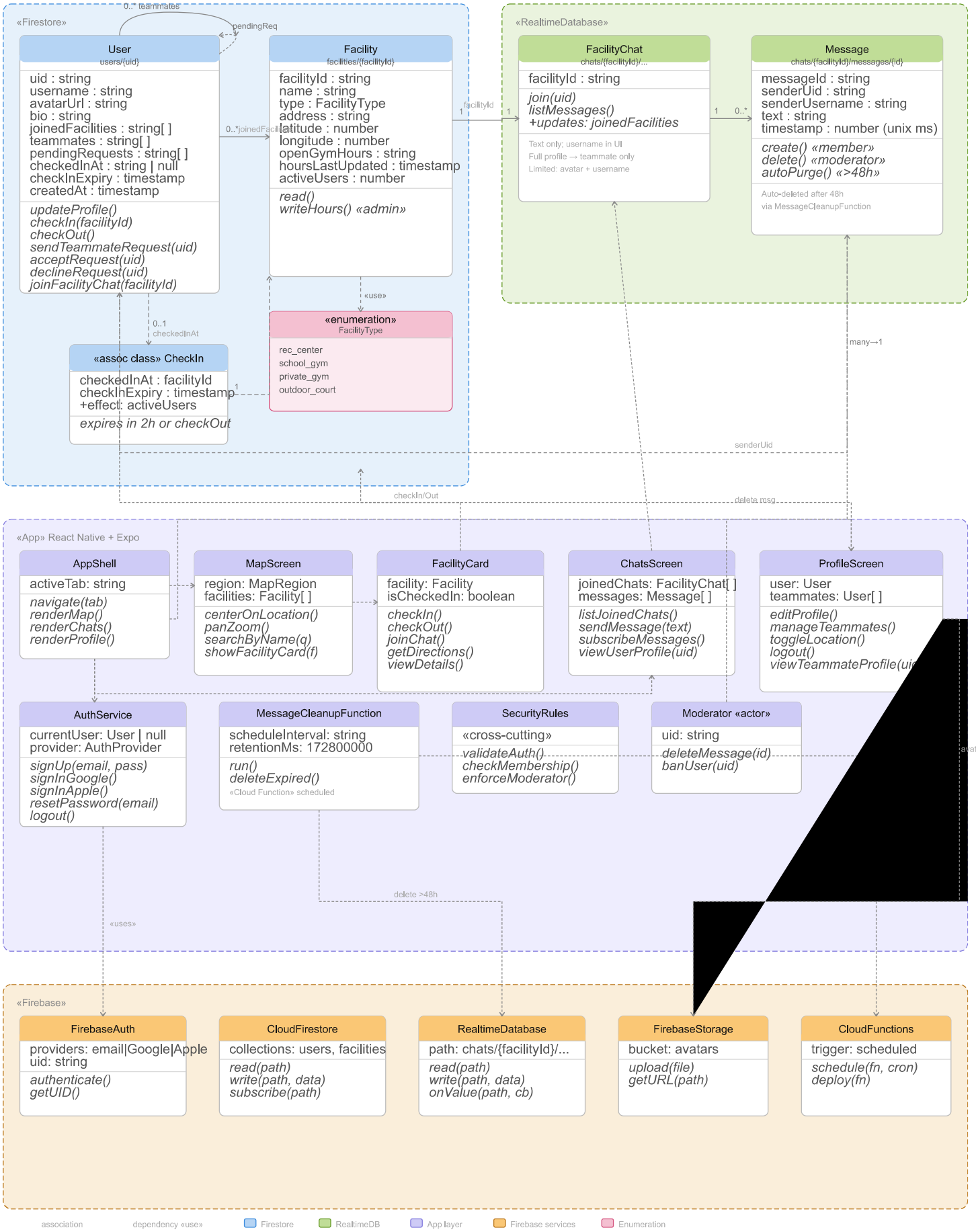
«Firestore»
0..* teammates
pendingReq
User
users/{uid}
uid : string
username : string
avatarUrl : string
bio : string
joinedFacilities : string[ ]
teammates : string[ ]
pendingRequests : string[ ]
checkedInAt : string | null
checkInExpiry : timestamp
createdAt : timestamp
updateProfile()
checkIn(facilityId)
checkOut()
sendTeammateRequest(uid)
acceptRequest(uid)
declineRequest(uid)
joinFacilityChat(facilityId)
0..*joinedFacilities
Facility
facilities/{facilityId}
facilityId : string
name : string
type : FacilityType
address : string
latitude : number
longitude : number
openGymHours : string
hoursLastUpdated : timestamp
activeUsers : number
read()
writeHours() «admin»
«use»
«enumeration»
FacilityType
rec_center
school_gym
private_gym
outdoor_court
0..1
checkedInAt
«assoc class» CheckIn
checkedInAt : facilityId
checkInExpiry : timestamp
+effect: activeUsers
expires in 2h or checkOut
1
«RealtimeDatabase»
FacilityChat
chats/{facilityId}/...
facilityId : string
join(uid)
listMessages()
+updates: joinedFacilities
Text only; username in UI
Full profile → teammate only
Limited: avatar + username
facilityId
1
1
1
0..*
Message
chats/{facilityId}/messages/{id}
messageId : string
senderUid : string
senderUsername : string
text : string
timestamp : number (unix ms)
create() «member»
delete() «moderator»
autoPurge() «>48h»
Auto-deleted after 48h
via MessageCleanupFunction
many→1
senderUid
checkIn/Out
delete msg
«App» React Native + Expo
AppShell
activeTab: string
navigate(tab)
renderMap()
renderChats()
renderProfile()
MapScreen
region: MapRegion
facilities: Facility[ ]
centerOnLocation()
panZoom()
searchByName(q)
showFacilityCard(f)
FacilityCard
facility: Facility
isCheckedIn: boolean
checkIn()
checkOut()
joinChat()
getDirections()
viewDetails()
ChatsScreen
joinedChats: FacilityChat[ ]
messages: Message[ ]
listJoinedChats()
sendMessage(text)
subscribeMessages()
viewUserProfile(uid)
ProfileScreen
user: User
teammates: User[ ]
editProfile()
manageTeammates()
toggleLocation()
logout()
viewTeammateProfile(ui
AuthService
currentUser: User | null
provider: AuthProvider
signUp(email, pass)
signInGoogle()
signInApple()
resetPassword(email)
logout()
MessageCleanupFunction
scheduleInterval: string
retentionMs: 172800000
run()
deleteExpired()
«Cloud Function» scheduled
SecurityRules
«cross-cutting»
validateAuth()
checkMembership()
enforceModerator()
Moderator «actor»
uid: string
deleteMessage(id)
banUser(uid)
avatar
delete >48h
«uses»
«Firebase»
FirebaseAuth
providers: email|Google|Apple
uid: string
authenticate()
getUID()
CloudFirestore
collections: users, facilities
read(path)
write(path, data)
subscribe(path)
RealtimeDatabase
path: chats/{facilityId}/...
read(path)
write(path, data)
onValue(path, cb)
FirebaseStorage
bucket: avatars
upload(file)
getURL(path)
CloudFunctions
trigger: scheduled
schedule(fn, cron)
deploy(fn)
association
dependency «use»
Firestore
RealtimeDB
App layer
Firebase services
Enumeration